

Design Document

Problem Overview :

Acme generates educational content like quizzes, lessons, rationale. Now, the thing is, these source documents are not static over time. The content may get updated. When that happens, the previously generated content may become stale or out of date.

So the goal is to build a self-healing content system that continuously monitors source material, detects changes, identifies affected learning content, proposes patch fixes and performs content updation via confidence scored automation or human review before publication.

Example workflow:

Source change detected (probably via cron trigger) → AI analyse semantic diff (trivial/cosmetic change or big update) → AI finds affected content via knowledge graph → AI generates updated proposal → Human / automatic approval based on confidence score → publish updated content

What entities exist?

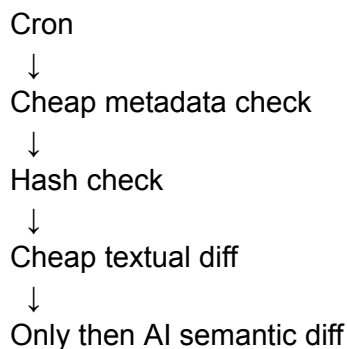
Schemas

- Sources
 - id
 - name
 - url
 - latest_hash
 - updated_at
- Concepts
 - id
 - name
 - category
 - updated_at
- Content
 - id
 - type
 - body
 - version
 - updated_at
- Change sets
 - Id
 - Updated_at
 - Source_id
 - Concept_id

- SourceSnapshots
 - id
 - source_id
 - content
 - hash
 - created_at
- PatchProposals
 - id
 - changeset_id
 - content_id
 - old_content
 - proposed_content
 - confidence_score
 - status
 - created_at

How is source change detected?

- A scheduled cron job periodically fetches source documents. Each document is normalized and hashed using SHA-256. If the new hash differs from the stored hash, the source is added to a daily ChangeSet.
- Instead of Cron check on every source, we should do:



How is semantic diff computed?

- Once a ChangeSet is created, AI compares the previous and current source snapshots to classify:
 - Added concepts
 - Removed concepts
 - Deprecated concepts
 - Severity of semantic change

How does KG traversal work?

- A knowledge graph is modeled relationally in postgresQL using concept nodes and mapping tables for the sake of simplicity.

How is impact severity scored?

- We can think of it like this: if the content update only contained minor CSS changes or some website modifications, then obviously the severity is very low, and this does not mean that we need to update our generated content.
- Only when the source change is big, like some deprecation of old sources or functions or documents, or some curriculum changes for some educational content, then only we can consider it as a high impact severity.

When is auto-healing allowed?

- Its allowed when the confidence score is high. Confidence score in itself is not just a simple AI-generated score. It should take into account all the different things, like how trivial the change is and some other aspects as well like:
 - Semantic ambiguity
 - Source citation quality
 - Patch complexity
 - etc

What does the reviewer UI show?

- Reviewer UI should show the diff similar to how we see in code changes, so we can see the old content and the updated content that the AI has updated based on its own analysis.
- We should be able to see the severity score and the confidence score and all the impacted content for the particular source concept.

Tech stack:

UI: ShadCN and TailwindCSS

FullStack: [Next.js](#), Typescript

DB : Postgres (Supabase)

AI: Gemini API

Agent orchestration: Vercel AI SDK (not agent heavy)

Non- Goals:

- Realtime source monitoring
- Full production scale system or graph db
- Auto publish without audit